# Mohamed Osama Zahran

Software Engineer

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## PROFESSIONAL SUMMARY

Software Engineer with 3+ years of production delivery — six parallel enterprise systems at RICOH Europe (technical design through implementation, testing, deployment and stabilization), a live multi-portal government healthcare application, and e-commerce products in production with real users and real payments. Ships complete features across the stack: React 19 + Next.js + TypeScript front ends, services and APIs in Node.js (NestJS) and ASP.NET Core, relational and non-relational databases (PostgreSQL, MongoDB, SQL Server), and third-party integrations (Stripe, Paymob, SharePoint, AS400, HuggingFace). Builds AI-powered systems for real: LangGraph multi-agent pipelines, LLM tool-calling with SSE streaming, moderation services live on HuggingFace Spaces, and curated-corpus retrieval with deterministic re-ranking — gated by automated suites (504 tests on the latest AI system, 409 on a freelance platform). Authored the team's Next.js frontend boilerplate (scaffolding, boundary guards, CI) and works AI-first with Claude Code and Codex in the daily loop — reviewing, understanding, and hardening every line that ships.

## TECHNICAL SKILLS

**Languages: TypeScript, JavaScript (ES6+), Python, SQL**

**Frontend (React ecosystem): React 19, Next.js 15/16, TypeScript, JavaScript (ES6+), HTML5/CSS3, e-commerce storefronts & dashboard applications, reusable component libraries & design systems (Shadcn / Radix UI), hooks & modern patterns, state management (Redux Toolkit, Zustand, TanStack Query/Form), REST API integration with secure JWT auth & RBAC, responsive & accessible interfaces (WCAG), React Router, modular & scalable front-end architecture, performance & bundle optimization, RTL + next-intl i18n**

**Backend & APIs (working knowledge): NestJS, ASP.NET Core, EF Core, MongoDB, SQL Server, PostgreSQL, Prisma, REST API contracts, SignalR/WebSockets**

**AI & Integrations:** LangGraph multi-agent pipelines, LLM tool-calling + SSE streaming, HuggingFace Transformers (Llama 3, Qwen 2.5, Whisper, DistilBERT), Google Gemini, Redis

**Styling & UI: Tailwind CSS v4, Shadcn / Radix UI design system, Ant Design, Framer Motion, GSAP, Lenis, RTL + next-intl i18n, WCAG accessibility**

**Tools & Practices: Git (pull requests, code reviews), GitHub Actions CI/CD with quality gates, Vitest (390+ tests), xUnit, Zod, Swagger/OpenAPI, Docker & docker-compose, Vercel + containerized cloud deployment, Lighthouse & performance profiling, technical-debt identification & sustainable remediation, code-standards ownership**

## PROFESSIONAL EXPERIENCE

**RICOH Europe** — Maadi, Cairo *Nov 2025 – Present*

*Full Stack Engineer*

- Own full-stack delivery across six production enterprise systems in parallel (CORELIA program, cross-regional European teams): technical design through implementation, testing, deployment and stabilization; set component patterns, code standards and review practices for the front-end surface.
- Collaborate daily with multicultural, cross-functional teams in a corporate European environment.
- Authored the team's Next.js frontend boilerplate adopted across CORELIA products — typed routes, feature scaffolding, Zod boundary guards, Vitest/Playwright setup, Docker behind Nginx via GitLab CI — and wrote the frontend architecture and API-reference docs for the surveillance system.
- Profile and optimize production front ends — bundle analysis, code splitting, virtualized tables and memoization keep data-heavy real-time dashboards smooth and responsive.
- Work AI-first with Claude Code and Codex coding agents: review, test, and validate AI-generated code, understand every line, and gate merges on CI test suites.

**NedSwiss — Remote, Switzerland Oct 2023 – Nov 2025**

*Full Stack Engineer*

- Built a multilingual admin/staff CRM dashboard (Next.js 15, TanStack Table/Query, Recharts, react-quill) with role-based navigation, rich data tables, chart analytics, and real-time cache invalidation.
- Developed the company marketing website with advanced scroll animations (GSAP, Lenis), next-intl i18n routing, Zod-validated lead capture, and Turbopack bundle optimization.

**GDSC – Shorouk Academy** — Shorouk City *Aug 2023 – Jun 2025*

*Frontend Lead*

- Led the frontend track of the Google Developer Student Club: designed and ran hands-on workshops in React, Next.js, and TypeScript, and coached developers through modern front-end architecture patterns.
- Mentored developers in React, TypeScript, state management, and modular UI architecture; authored workshop material reused across cohorts.

## PROJECTS

**Tailgating Detection System — Real-Time Surveillance Dashboard (Nov 2025 – Present)**

*React 18, TypeScript, Vite, Redux Toolkit, TanStack Query, Zod, Shadcn UI, MJPEG*

- Real-time anomaly dashboard: live MJPEG channel streaming, license-plate OCR workflows, anomaly and history views, role-based access, and audit trails for physical-security operations.
- JWT authentication with token refresh wired across three REST APIs (surveillance v6, license-plate OCR, MOT) with OpenAPI-documented contracts; authored the system's frontend architecture and API-reference documentation.

**3D Building & Space Viewer — Digital-Twin Capture Explorer**

*React 19, TypeScript, React Three Fiber + Drei (Three.js), Redux Toolkit, Tailwind CSS, Framer Motion, Radix UI*

- Multi-mode viewer with dedicated Dollhouse / Panorama / Split layouts: textured mesh and 360 imagery side by side, room-aware navigation, and door-constrained markers generated from transform metadata.
- Progressive panorama loading with preload cues and animated transition overlays, keeping navigation fluid on capture datasets of unbounded size.
- Data-driven scene setup: camera transforms and room assignments load from JSON, so new building capture runs go live without code changes.

**AGL Invoice Management — Enterprise Invoice Processing (SharePoint → AS400)**

*React, TypeScript, React Hook Form + Zod, Axios, shadcn/ui*

- Invoice-processing pipeline: scanned invoices downloaded from SharePoint (or direct upload), OCR extraction of fields and line items, then a side-by-side document-vs-extracted-data verification UI (MultiDocEditor) before validated auto-submission to the enterprise AS400 API with legacy CFFFRE formatting.
- Feature-driven architecture (features/invoices/{api, components, hooks, types, utils, constants}) with SharePointModal and CFFFRE formatter utilities isolated for reuse.

**Qalam Vision — Arabic NLP Annotation Interface**

*React, TypeScript, Label Studio integration*

- Arabic/multilingual keyword-extraction interface integrating Label Studio annotation workflows for the NLP team's data pipeline.

**RicohFacturation — Supplier Invoice Control Platform (2026)**

*Blazor WebAssembly, ASP.NET Core, PostgreSQL, Docker Compose, MudBlazor, Tailwind v4 design tokens*

- Invoice control against contracts, meters and prestations referentials: period imports, conformity tracking per control family, credit-note (avoirs) generation — French-first UI, every string i18n-keyed.
- Microsoft Entra optional SSO, transactionally audited user administration, component test suites (culture, pagination, status gating), and Dockerized dev/prod deployment via Compose.

**CORELIA Next Boilerplate — Team Frontend Platform (authored)**

*Next.js 16, React 19, TypeScript strict, Tailwind v4, Zod 4, Vitest + Playwright, Docker + Nginx via GitLab CI*

- Authored the team's base for product frontends: App Router typed routes, React Compiler, feature scaffolder CLI (npm run corelia -- feature), and Zod guards at env, form, and risky-API boundaries.
- Ships configured Vitest + Testing Library, per-project Playwright, an AGENTS.md + agent-playbook with a single verify gate before handoff, and Docker-behind-Nginx deployment through GitLab CI.

**OZ Uniform Store** — Full-Stack Client Product *(Jul 2026 – Aug 2026)*

*Next.js 16, React 19, Tailwind CSS v4, TanStack Form + Zod 4, .NET 10 Web API, EF Core 10, SQL Server, Hangfire*

- Arabic-first RTL e-commerce storefront for a school-uniform retailer: school/grade catalog, cart, cash-on-delivery checkout, Bosta courier shipping, and per-order token tracking URLs.

- Built the .NET 10 backend: JWT auth, Hangfire background jobs (auto-cancel orders, email), FluentValidation, Bosta shipping integration, and audit logging; deployed to production.

### Medical System — Gov Healthcare Insurance Portal (Aug 2026)

*Next.js 16, React 19, TypeScript, TanStack Query, React Hook Form + Zod, next-intl (AR/EN RTL), Recharts, Shadcn / Radix UI, Tailwind CSS, Biome, Vitest*

- Co-built a production government healthcare portal (citizen, doctor, and admin portals, live on Vercel) with a 2-developer team, contributing the majority of commits (40 of 59): the full admin verification workflow across 4 phases (application review queue, card lifecycle management, category configuration, verification workbench).
- Built the doctor clinical workspace (point-of-care desk with patient chart and visit management) and the citizen insurance experience (application, tracking history, card-issuance stepper) on OpenAPI-snapshot API contracts with role-based auth.

### ARME Egypt (Arab River Medical) — B2B Medical Equipment Portal (May 2026 – Jun 2026)

*Next.js 16, React 19, TypeScript, Tailwind CSS v4, next-intl (EN/AR RTL), Docker*

- Bilingual English/Arabic (RTL) B2B portal for a surgical-instruments and medical-furniture supplier: public marketing site with context-aware lead capture that geo-routes Egyptian visitors to a catalog-download flow and international visitors to an agency-inquiry flow.
- Secure admin dashboard for catalogs, lead review, and analytics; JWT session handling, honeypot anti-spam, XSS-safe JSON-LD, WCAG AA contrast, and a dynamic sitemap; live in production.

### Wasla CRM — Multi-Tenant SaaS Platform (Stripe & Paymob Payments) (Nov 2025 – Jun 2026)

*Next.js 15, ASP.NET Core 8, FastAPI, SignalR, Hangfire, Stripe, Paymob, HuggingFace*

- Full CRM platform spanning frontend, .NET 8 backend, and two AI microservices: multi-tenant JWT auth with per-user permissions, SignalR real-time notifications, Hangfire scheduled jobs
- **AI layer**: Llama-3.3/Qwen tool-calling backend, and **toxicity moderation** (DistilBERT / DeHateBERT) with sub-100ms latency on high-confidence predictions.

### Nabdh Platform — Home Nursing Marketplace Backend (Jun 2026 – Present)

*NestJS 10, TypeScript, MongoDB 7 (2dsphere geospatial), WebSocket, MinIO S3, Docker*

- Engineered the modular-monolith backend for an Egyptian home-nursing marketplace: auth, booking, payment, chat, and notification modules serving patient and nurse Flutter apps as well as a React admin panel.
- Geospatial matching with MongoDB 2dsphere indexes, real-time WebSocket chat, and S3-compatible media storage via MinIO.

**CyrusLearn** — Interactive Learning Platform (WE Innovative Hackathon) *(Mar 2026 – Apr 2026)*

*React 19, Vite, TypeScript, React Three Fiber, Redux Toolkit, Shadcn UI*

- Feature-driven e-learning platform with 3D interactive labs, structured learning paths, a quiz engine, collaborative rooms, Google OAuth, subscription flow, and an admin dashboard.

### MAC-Egypt — UNIDO/NOU/EED Assessment & Monitoring Platform (Oct 2026 – Present)

*Next.js 16, React 19, TypeScript, Tailwind CSS v4, shadcn/ui, next-intl (Arabic-first RTL)*

- Freelance build for a UNIDO/NOU/EED program: workshop registry with duplicate detection, a 13-section field survey wizard with GPS and photo evidence, machine-scored selection runs, training and equipment-delivery records, and role-shaped dashboards.
- Bilingual Arabic-first RTL/English UX on a schema-driven survey engine; 409 passing tests gating a GitHub Actions deploy pipeline shipping to production.

### Meapal LogBook — Activity Platform for RICOH/Corelia Teams (Sep 2026)

*React 19, Vite, TypeScript, TanStack Query, Tailwind CSS v4, Zod 4, shadcn/ui, Sonner*

- Daily activity log for RICOH/Corelia teams: calendar-driven pay-period tracking against the 160h period target with a pace marker showing where you should be by today, stretch-target pacing, and weekly charts that open any day they represent.
- Work streaks honoring Fri/Sat rest days (never break the streak, working one counts it), personal task records raced against your own best day, and reminders that stay quiet on rest days.
- Three role surfaces (employee, project manager, admin): employees log and track, project managers get team analytics and reports, admins administer projects and users.
- Optimistic-everything UX: instant submits/edits/deletes with 5-second undo-toast windows, localStorage autosave with crash-recovery prompts, and per-browser draft resumption — no global spinners, no confirm dialogs.

### agentic-shop — Agentic AI Shopping System (Aug 2026)

*Python, FastAPI, LangGraph, Pydantic v2, Next.js 16, React 19, SSE*

- Built a multi-agent shopping pipeline (LangGraph) that turns a one-sentence goal into ranked products, a conversational answer, and a validated UI plan rendered client-side over SSE streaming.
- Deterministic product scorer with Pydantic v2 contracts; fully offline mock mode with 504 passing tests; real-LLM swap-in ready.

### Attendance API — GPS-Verified Workforce Service (Aug 2026)

*NestJS 11, TypeScript, MongoDB (Mongoose 9), JWT + Passport, Swagger/OpenAPI, class-validator*

- Built a NestJS service for GPS-verified employee attendance: per-workplace geofencing, timezone-aware working days, and a calendar history that classifies each day as present, incomplete, absent, weekend, or upcoming.
- JWT authentication with bcrypt hashing, rate-limited endpoints, OpenAPI/Swagger documentation, and a published Postman collection.

**Gym Management API (Aug 2026)**

*ASP.NET Core 8, EF Core, SQL Server, xUnit, Swagger/OpenAPI*

- Single-tenant gym subscription API with cryptographic QR check-in: atomic session deduction under concurrency (SQL applocks + conditional updates), preventing negative balances.
- Time-based and per-sale session-credit subscriptions, RFC 7807 errors with machine-readable reason codes, structured logging, health checks, and an xUnit test suite.

**NED Swiss — Digital Agency Website (Oct 2023 – Nov 2025)**

*Next.js 15, React 19, TypeScript, Tailwind CSS v4, next-intl (EN/DE/FR), GSAP, Framer Motion, Lenis*

- Trilingual (English/German/French) marketing website for a Swiss digital agency: locale routing, GSAP scroll animations, Lenis smooth scrolling, magnetic hover micro-interactions, and per-locale SEO metadata; in production at ned-swiss.ch.

**NED Swiss — Admin/Staff CRM Dashboard (Oct 2023 – Nov 2025)**

*Next.js 15, TypeScript, TanStack Table + TanStack Query, Recharts, react-quill*

- Multilingual admin/staff CRM dashboard: rich data tables with server-side mutations, chart analytics, rich-text editing, and real-time data sync across Arabic and English locales.

**AI-Powered Translation Platform — Freelance (2024)**

*Next.js, TypeScript, Tailwind CSS, next-intl, Paymob*

- Multilingual translation product (Arabic/English/French) with end-to-end Paymob billing, a daily-quota credit system, admin transaction monitoring, and a blog module.

## EDUCATION

**BSc in Computer Science** — Shorouk Academy, Cairo *GPA: 3.8/4.0*